Adoption of electric vehicle: A literature review

Abstract

Nowadays, increasing environmental concerns and the depletion of fossil fuel resources have brought Electric Vehicles (EVs) to the forefront as a sustainable alternative in the global transportation sector. The adoption of EVs plays a pivotal role in the global transition to sustainable transportation. This study primarily focuses on the factors influencing consumer behavior in the adoption of EVs, particularly in the Indian context. This comprehensive review of a total of 42 studies highlights the challenges, barriers, and opportunities in the adoption of EVs. Despite various efforts by governments, industries, and researchers, EV adoption continues to face numerous challenges. This paper compiles and summarizes existing literature to identify the key barriers, motivators, and strategies for enhancing EV acceptance. A detailed analysis is presented on the factors influencing EV adoption, including infrastructure, consumer behavior, cost, policy frameworks, and technological advancements. The literature is categorized based on major themes related to EV adoption, such as Challenges and Barriers to EV Adoption, Consumer Behavior and Purchase Intentions, Government Policies and Incentives, Technological and Market Development, and Insights from International Markets. Furthermore, several studies highlight the critical role of technological advancements in the automobile industry, such as improved battery performance and fast-charging systems, in addressing consumer concerns and enhancing the usability of EVs. The findings emphasize the need for improved charging infrastructure, affordable pricing, and consumer awareness campaigns. This work aims to provide insights into current research gaps and future directions to promote the EV ecosystem. As per the conclusion of this study, future research should focus on integrating renewable energy sources with EV infrastructure, understanding regional disparities in adoption rates, and enhancing battery recycling technologies. Furthermore, it explores the future scope for the EV market, including infrastructure development, battery technology improvements, and consumer education.

Key Words: *Electric Vehicles (EVs), EV Adoption, EV Infrastructure, EV Ecosystem, EV Challenges etc.*

Introduction

As everyone knows, air pollution is a high-priority threat in the global context. The electric vehicle (EV) market is undergoing significant transformation due to increasing concerns about environmental sustainability and the need to reduce carbon emissions from the environment. This paper seeks to analyze the key factors influencing customer buying behavior in the automobile market, based on previous literature and the current requirements to improve the adoption of EVs in India. It also helped to raise interest in green technologies and sustainable transportation, understanding consumer behavior is essential for shaping policies and market strategies aimed at promoting EVs. The EVs cannot run without charging batteries so it is very important to establish the required infrastructure for the charging stations.

Review of Literature

Topic	Authors & Year	Focus/Key Points
1. Challenges in EV Adoption	P. Muthulakshmi et al., 2023	Obstacles like power supply shortages, fewer charging stations, charging time, service, costs, and battery issues.
	Fayez Alanazi, 2023	Challenges such as cost, infrastructure, charging stations, range limitations, and battery performance.
	Anthony Anosike et al., 2023	Challenges in fleet size, capacity, and infrastructure for final-mile parcel delivery.

	Ramniwas Singh et al., 2023	Inadequate infrastructure, lack of new tech, unskilled workforce, policy frameworks, and manufacturing concerns.
	Ashish Sethiya, 2018	Challenges such as cost, infrastructure, charging facilities, and reliability of EVs.
2. Perception & Behavior Towards EVs	Sonali Goel et al., 2021	EV types, approaches, and charging infrastructure as barriers to adoption.
	G. Krishna, 2021	Public perception and barriers to adoption using thematic analysis, factors affecting EV purchase.
	Tejesh et al., 2023	Consumer behavior, environmental concerns, government policies, and price influencing EV adoption.
	Priyanka Ranawat et al., 2023	Consumer attitudes towards EVs, range anxiety, policy support, and infrastructure availability.
	Nitin J. et al., 2022	Factors like price, infrastructure, environmental concerns, and knowledge; mediating role of government policies.
3. Consumer Behavior & Decision Factors	Priyanshu Kumar et al., 2022	Consumer behavior using YouTube reviews and sentiment analysis, revealing positive feedback and consumer confusion.
	Channabasav Hiremath et al., 2022	Factors like government incentives, technology, speed, and price influencing EV purchase decisions.
	Silvana Secinaro et al., 2022	Consumer behavior insights using bibliometric and thematic analysis, consumer characteristics and methodologies.
	Mr. Omkar Tupe et al., 2023	Consumer perceptions based on vehicle choice, gender, and income, importance of infrastructure for EV adoption.
4. Infrastructure, Policy & Technological	Udit Chawla et al., 2023	Public perception on autonomous driving, charging time, innovation, and affordability.
	H. Nurgul DURMUS SENYAPAR et al., 2023	Policies and incentives for EV adoption across countries, reducing carbon footprints.
	Vedant Singh et al., 2021	EV policy and infrastructure development in India, research funding needed for charging stations and EVs.
	Fei Ye et al., 2024	Psychological factors and policy attributes (subsidies, control) in influencing EV adoption.
5. Statistical Models & Data Analysis	Jingnan Zhang et al., 2022	Structural Equation Modeling comparing adoption factors in cities, impact of subsidies.
	G. Mopi et al., 2023	Discrete choice models to analyze EV adoption factors in Greece (financial incentives, environmental awareness, infrastructure).

	Dutta B. et al., 2021	Modified Theory of Planned Behavior (TPB) to understand consumer intentions toward sustainable EV adoption.
	Mishra, S. & Malhotra, G., 2019	Confirmatory factor analysis on factors driving EV adoption in India (environmental concerns, vehicle performance).
6. Market Trends & Future Scope	Sitendu Roy et al., 2022	Factors influencing decisions, research gaps, strategies for EV manufacturers.
	Priyanka Ranawat et al., 2023	Overview of the EV market in India, government policies, sustainability, and EV developments.
	Henry Lee et al., 2018	Need for reducing EV costs and expanding charging infrastructure for large-scale adoption.
	Kumari, S. et al., 2021	Comparison of EV adoption in India and Norway, technological and policy factors influencing growth.
7. Psychological & Socio-Psychological	Malik, C.et al., 2017	Model on the relationship between environmental attitudes and purchasing intentions for eco-friendly products.
	Kim, N. et al., 2013	Impact of eco-labels and ease of purchase on eco-friendly product behavior.
	Upadhyaya, M., 2017	Environmental familiarity and subjective norms influencing eco-friendly vehicle purchases.
	Bhatia Mayank et al., 2023	Impact of awareness, green values, and perceptions of companies' commitment to green marketing on consumer preferences for green products.
8. EV Adoption in Developing vs Developed	Knez, M.et al., 2017	Factors influencing purchasing decisions for low-emission vehicles in Slovenia (financial, technical, environmental).
	Ram, A. 2020	Deterrents to EV adoption (price, infrastructure, driving range, charging time).
	Jayasingh, S et al., 2021	Factors influencing electric two-wheeler adoption in India, with perceived economic benefits as a major driver.
9. Electric Vehicle Lifecycle & Sustainability	Sharma, A. & Joshi, M., 2022	Lifecycle analysis of EVs, challenges in battery disposal, long-term environmental impact, and sustainability concerns.
	Robinson, K. et al., 2020	Environmental and social impacts of EVs, including the carbon footprint reduction potential.
10. Technological Innovations in EVs	Deshmukh, A. & Gohil, H., 2023	Impact of recent technological advancements in battery efficiency and charging speed on EV adoption rates.
	Sharma, R. et al., 2022	Advances in AI and machine learning for smart charging stations, autonomous vehicles, and data-driven EV services.

11. Government & Corporate Role in EV Adoption	Reddy, P. et al., 2021	Government incentives, policy frameworks, and corporate responsibility in driving EV adoption.
	Gupta, D. et al., 2021	Corporate social responsibility (CSR) initiatives by EV manufacturers and their influence on customer adoption behavior.
12. Environmental & Social Impact of EVs	Singh, A. et al., 2021	Social and environmental benefits of EV adoption, including reduced air pollution, lower noise pollution, and improved urban mobility.
	Verma, S. et al., 2021	A deep dive into the societal shifts required for large-scale EV adoption, including infrastructure, policies, and behavior changes.

Data Analysis

Challenges in EV Adoption

- ✓ Infrastructure Issues: Lack of charging stations and slow charging times are major challenges to EV adoption.
- ✓ High Initial Cost: The high upfront cost of EVs, primarily due to battery technology, remains a barrier.
- ✓ Lack of Skilled Workforce and Technology: The lack of qualified personnel and inadequate technological infrastructure are hindering EV adoption.

Consumer Perception and Behavior

- ✓ Range Anxiety: Concerns about the limited driving range of EVs affect consumer willingness to switch from internal combustion engine (ICE) vehicles.
- ✓ Environmental Awareness: Growing concerns over climate change and the environmental benefits of EVs have increased consumer interest.
- ✓ Cost and Affordability: While EVs offer long-term savings in maintenance and fuel costs, the upfront price remains a significant hurdle for many consumers.

Role of Policy and Government

- ✓ Government Incentives: Tax breaks, subsidies, and infrastructure development are critical to encouraging EV adoption.
- ✓ Policy Frameworks: Government regulations for EV standards and battery disposal are necessary to support long-term growth.

Technological Innovations

- ✓ Battery Efficiency and Charging Speed: Advancements in battery technology and fast-charging systems are reducing consumer concerns about range and charging times.
- ✓ Smart Charging and Autonomous Vehicles: AI-powered smart charging stations and autonomous driving technologies are poised to revolutionize the market.

Market and Consumer Trends

- ✓ EV Market Growth: Consumer interest in EVs is growing, but adoption is still heavily influenced by government policies and infrastructure development.
- ✓ Corporate and CSR Initiatives: Companies are integrating sustainability initiatives to promote EV adoption and influence consumer behavior.

Environmental and Social Impact

- ✓ Reduction in Carbon Footprint: EVs offer substantial reductions in emissions and air pollution, benefiting urban environments.
- ✓ Social Responsibility: Adoption aligns with the global push for reducing dependence on fossil fuels and promoting cleaner transportation.

Key Findings

1) Infrastructure, high initial costs, and technology gaps remain the primary barriers to EV adoption.
2) Range anxiety, environmental consciousness, and upfront costs are key factors influencing EV purchasing decisions.
3) Government incentives, subsidies, and infrastructure development play a critical role in shaping EV adoption.
4) Battery and charging technology improvements are essential for increasing the practicality of EVs.
5) The EV market is growing, driven by both environmental awareness and government policies, with significant potential for expansion.
6) EV adoption helps reduce the carbon footprint and contribute to sustainability goals.

Future Scope

1) Infrastructure in Urban and Rural areas: Expanding and Developing ultra-fast charging stations in underserved areas is crucial for improving EV adoption.
2) Sustainable Battery Recycling: advancements in battery technology addressing the environmental impact of used batteries will become increasingly important.
3) Policies and International Collaboration: A global approach to policy regulation will create consistency in EV adoption across countries as well as international collaboration should collaborate to accelerate infrastructure and technology development.
4) Technological Integration in EVs: Integration of AI and autonomous technology into EVs will further transform the market and the development of a smart city ecosystem will improve efficiency
5) Sustainability and Circular Economy: Increasing emphasis on recycling and repurposing EV parts and batteries will contribute to sustainability goals.

Conclusion

The adoption of electric vehicles has a significant opportunities and challenges in India as per the study. While Indian government policies and technological advancements are paving the way for a sustainable future and with the help of this literatures are addressing infrastructure issues, improving battery technology, and enhancing consumer awareness are critical to fostering widespread adoption. In future the adoption of EVs depends on continued innovation, policy support, and global collaboration to overcome the barriers to adoption and fully realize the environmental and economic benefits of electric transportation as per the data analysis of previous literatures.

References

P.Muthulakshmi, T. Tamilarasi, Tanmay Tapan Banerji, S Albert Antony Raj & E Aarthi (2023). Impact and Challenges to Adopting Electric Vehicles in developing countries – a case study in India, EAI Endorsed Transactions on Energy Web, doi: 10.4108/ew.2665, 1-8.

Mr. Tejesh, Dr. Susheela Devi B Devaru (2023), Consumer Buying Behaviour Towards Electric Vehicles in Bengaluru City, Karnataka, India, IJFMR, E-ISSN: 2582-2160, Volume 5, Issue 5, September-October 2023, pp 1-9.

Alanazi, F (2023). Electric Vehicles: Benefits, Challenges, and Potential Solutions for Widespread Adaptation. Appl. Sci. 2023, 13, 6016.

Anthony Anosike, Helena Loomes, Chinonso Kenneth Udokporo & Jose Arturo Garza-Reyes (2023) Exploring the challenges of electric vehicle adoption in final mile parcel delivery, International Journal of Logistics Research and Applications, 26:6, 683-707, DOI: 10.1080/13675567.2021.1978409

Ramniwas Singh & Arvind Kumar (2023). Comprehensive Analysis of Challenges Hindering the Adoption of Electric Vehicles in India, International Journal of Science and Research (IJSR), Volume 12 Issue 6, June 2023, 1768-1770

Ashish Sethiya (2018). ELECTRIC VEHICLES IN INDIA: CHALLENGES & OPPORTUNITIES, JETIR October 2018, Volume 5, Issue 10, ISSN-2349-5162,699-702.

Sonali Goel, Renu Sharma & Akshay Kumar Rathore (2021). A review on barrier and challenges of electric vehicle in India and vehicle to grid optimization, Transportation Engineering 4 (2021) 100057, 1-14.

G. Krishna (2021). Understanding and identifying barriers to electric vehicle adoption through thematic analysis, Transportation Research Interdisciplinary Perspectives 10 (2021) 100364, 1-9.

Dr. Priyanka Ranawat, Sashant Sharma, Mr. Suraj Kumar (2023). Study On Market Sustainability and Consumer Behaviour Towards Electric Vehicles (Ev) In India, IJIRA, ISSN:2583-0295, Volume 03, No. 02(II), April- June 2023, pp 48-54.

Ishika Ranjan, Sawan Kumar Jha, Shuvam Mondal (2022). A study on Consumer Buying Behaviour towards Electric Vehicle, ISSN No:-2456-2165, Volume 7, Issue 6, June-2022, pp 325-346.

Priyanshu Kumar, Shahil Alam, Prof. Dr. Avinash Rana (2022). Consumer Behaviour on Electric Vehicles, IRJMETS, Volume:04/Issue:03/March-2022, pp 897-906.

Mr. Channabasav Hiremath, Dr. Rajeshwari B. Tapashetti, Dr. Rashmi R. Hunnur (2022), A Study on Consumer Buying Behaviour towards Electric Vehicles in Bagalkot, JETIR, ISSN-2349-5162, September 2022, Volume 9, Issue 9, pp b254- b278.

Silvana Secinaro, Davide Calandra, Federico Lanzalonga, Alberto Ferraris (2022), Electric vehicles consumer behaviours: Mapping the field and providing a research agenda, JETIR, ISSN-2349-5162, September 2022, Volume 9, Issue 9, pp b254- b278.

Mr. Omkar Tupe, P. S. (2020). CONSUMER PERCEPTION OF ELECTRIC VEHICLES IN INDIA. European Journal of Molecular & Clinical Medicine, 7(8), 4861-4869.

Chawla, U.; Mohnot, R.;Mishra, V.; Singh, H.V.; Singh, A.K (2023). Factors Influencing Customer Preference and Adoption of Electric Vehicles in India: A Journey towards More Sustainable Transportation. Sustainability, 15, 7020.

H. Nurgul DURMUS SENYAPAR, Murat AKIL (2023). Analysis of Consumer Behavior towards Electric Vehicles: Intentions, Concerns and Policies. Gazi University Journal of Science, Part C,11(1):161-183.

Vedant Singh, V. (2021). Analysis of electric vehicle trends, development and policies in India. Case Studies on Transport Policy, 9, 1180–1197. doi:https://doi.org/10.1016/j.cstp.2021.06.006

Fei Ye, W. K. (2021). Why do consumers choose to buy electric vehicles? A paired data. Transportation Research Part A, 147, 14-27. doi:https://doi.org/10.1016/j.tra.2021.02.014

Zhang, J.; Xu, S.; He, Z.; Li, C.; Meng, X(2022). Factors Influencing Adoption Intention for Electric Vehicles under a Subsidy Deduction: From Different City-Level Perspectives, 14, 1-24. https://doi.org/10.3390/su14105777.

Mishra, S. and Malhotra, G. (2019) Is India Ready for e-Mobility? An Exploratory Study to Understand e-Vehicles Purchase Intention. Theoretical Economics Letters, 9, 376-391. https://doi.org/10.4236/tel.2019.92027

Sitendu Roy & Dr Arvind Kumar Jain (2022). Consumer Adoption & Buying Behaviour For Electric Vehicles In India, International Journal of Mechanical Engineering, 7(5), 0974-5823.

Kumari Santosh, Sud Sameexa, Agarwal Anshula (2021). Fosterage and Sustenance of Electric Vehicles: A Comparative Study of India and Norway. SMS Journal of Entrepreneurship & Innovation. 2021; 8(1): 27 – 38. DOI: 10.21844/smsjei.v8i01.30003

Chirag Malik & Neeraj Singhal (2017). Consumer Environmental Attitude and Willingness to Purchase Environmentally Friendly Products: An SEM Approach, SAGE Publications, 21(2), 1-10, DOI: 10.1177/097226291770099

Nayeon Kim, Kyungtag Lee, Paul B. Tchounwou (2023). Environmental Consciousness, Purchase Intention, and Actual Purchase Behaviorof Eco-Friendly Products: The Moderating Impact of Situational Context, 20(7), doi:10.3390/ijerph20075312

Makarand Upadhyaya (2017). Purchase Behaviour of Indian Customers Towards Environment, International Journal of Mechanical and Production Engineering Research and Development (IJMPERD), 7(6), 109-120. 2249-6890

Bhatia, Mayank & Jain, Amit (2013). Green Marketing: A Study of Consumer Perception and Preferences in India, Electronic Green Journal, 1(36). DOI:10.5070/G313618392

Knez, M., Obrecht, M.(2017). Policies for Promotion of Electric Vehicles and Factors Influencing Consumers' Purchasing Decisions of Low Emission Vehicles, J. sustain. dev. energy water environ. syst., 5(2), pp 151-162. DOI: http://dx.doi.org/10.13044/j.sdewes.d5.0139

Aakarsh Ram (2020). Consumer Behavior towards Buying Electric, International Journal for Research in Applied Science & Engineering Technology (IJRASET), 8(5), 2321-9653. http://doi.org/10.22214/ijraset.2020.5098

Jayasingh, S.; Girija, T. (2021). Arunkumar, S. Factors Influencing Consumers' Purchase Intention towards Electric Two-Wheelers. Sustainability, 132, 2851. https://doi.org/10.3390/su132212851

G. Mpoi, C. Milioti, and L. Mitropoulos (2023). Factors and incentives that affect electric vehicle adoption in Greece, International Journal of Transportation Science and Technology, https://doi.org/10.1016/j.ijtst.2023.01.002

Henry Lee and Alex Clark (2018) Charging the Future: Challenges and Opportunities for Electric Vehicle Adoption, Belfer Center for Science and International Affairs Harvard Kennedy School, RWP18-026,1-77.

Dutta, B.; Hwang, H.-G (2021). Consumers Purchase Intentions of Green Electric Vehicles: The Influence of Consumers Technological and Environmental Considerations, Sustainability , 13, 12025.